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Permutation Codes in Levenshtein, Ulam and Generalized Kendall-tau Metrics

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***Abstract*—Our main goal in this work is to study permutation codes in the Levenshtein metric which can correct multiple deletions. Permutation codes in the Levenshtein metric are known to have a close relationship with those in Ulam and generalized Kendall-tau metrics. In this work, we present a new connection between different kinds of distances over two permutations, including Hamming, Levenshtein, Ulam, and generalized Kendall-tau distance. From this new connection and some known permutation codes in the Hamming metric, we obtain new permutation codes in Levenshtein, Ulam, and generalized Kendall-tau metrics with better size. In particular, we show that there exist permutation codes of length n for correcting t deletions with at most $(3t+1)\log n+o(\log n)$ bits of redundancy. Furthermore, we present the construction of our permutation codes for correcting t deletions with a specific decoding process.**

I. Introduction

Permutation codes, originally introduced by Slepian [1] in 1965 for transmitting data in the presence of additive Gaussian noises, have been extensively studied for their effectiveness in powerline transmission systems against impulsive noise [2], as well as in the development of block ciphers [3]. Recently, permutation codes have gained increased attention for their applications in flash memory technologies [4]–[7].

Flash memories have emerged as a promising nonvolatile data storage solution, favored for their speed, low power usage, and reliability. To circumvent the precise programming of each cell to a specific level in flash memory devices, Jiang et al. [4] introduced a rank modulation method that utilizes permutations to represent information. To address the issue of deletions or erasures in this context, Gabrys et al. [8] categorized these deletions into two types: symbol-invariant deletions (SID) and permutation-invariant deletions (PID). For example, $(1,\cancel{2},\cancel{3},4) \to (1,4)$ is a SID scheme, while $(1,\cancel{2},\cancel{3},4) \to (1,2)$ is a PID scheme.

Our goal in this work is to study permutation codes in the Levenshtein metric which can correct t deletions under the SID model, where deleting some symbols does not affect the values of others. For simplicity of expression, all *deletions* we refer to in the remainder of this paper represent *symbol invariant deletions*. Although there has been a significant breakthrough in the codes for correcting t deletions in general binary [9], [10] and non-binary [11] scheme, permutation codes for correcting t deletions are not as well-studied. Gabrys et al. [8] proposed permutation codes for correcting t deletions under the SID model by demonstrating the equivalence with permutation-invariant erasures (PIE) codes, which is facilitated through the integration of a permutation code in the Ulam metric. Besides, there is another line of work on studying permutation codes for correcting a burst of t deletions [12]–[14]. Order-optimal permutation codes capable of correcting a burst of at most t deletions are proposed in [12], [13].

In this paper, we start by establishing relationships between various kinds of distances over two permutations, including Hamming, Levenshtein, Ulam, and (generalized) Kendall-tau distance. Especially, the main idea of our work is that we introduce a novel mapping function such that we successfully build a tighter inequality bound between Ulam distance and Hamming distance compared with the result in [6]. It helps us to construct permutation codes for deletions/translocations/transpositions by applying the well-studied permutation codes in the Hamming metric for substitutions as the base code. In particular, we show that there exist permutation codes of length n for correcting t deletions with at most $(3t+1)\log n+o(\log n)$ bits of redundancy, where t is a constant.

Furthermore, we present the construction of permutation codes for correcting up to t deletions with a specific decoding process. Instead of relying on the auxiliary codes in the Ulam metric [8], our construction is achieved by incorporating the base code in the Hamming metric. The redundancy of our proposed code improves that of in [8], where t is a constant.

II. Notation and Preliminaries

Let $\Sigma_q=\{0,1,\ldots,q-1\}$ denote a finite alphabet of size q and Σ_q^n represents the set of all sequences of length n over Σ_q. We use bold letters to denote sequences, such as $\boldsymbol{u}$ and their elements with plain letters, e.g., $\boldsymbol{u}=u_1\cdots u_n$ for $\boldsymbol{u}\in\Sigma_q^n$. For functions, if the output is a sequence, we also write them with bold letters, such as $\boldsymbol{p}(\boldsymbol{u})$. The ith position in $\boldsymbol{p}(\boldsymbol{u})$ is denoted $p(\boldsymbol{u})_i$. Also, $\boldsymbol{u}_{[i,j]}$ denotes the substring beginning at index i and ending at index j, inclusive.

For ease of notation, we will denote the set $\{1,2,\ldots,m\}$ as $[m]$. Let n be a positive integer and $\mathcal{S}_n$ be the set of all permutations on the set $[n]$. Denote $\boldsymbol{\pi}=(\pi_1,\pi_2,\ldots,\pi_n)\in\mathcal{S}_n$ as a permutation with length n. Given a permutation $\boldsymbol{\pi}=(\pi_1,\pi_2,\ldots,\pi_n)\in\mathcal{S}_n$, the inverse permutation is denoted as $\boldsymbol{\pi}^{-1}=(\pi_1^{-1},\pi_2^{-1},\ldots,\pi_n^{-1})$. Here, π_i^{-1} indicates the position of the element i in the permutation $\boldsymbol{\pi}$.

Example 1. *Suppose* $\boldsymbol{\pi} = (1,3,4,2,5)$*, we have* $\boldsymbol{\pi}^{-1} = (1,4,2,3,5)$.

A. Basic Definitions

Definition 1. *For distinct* $i, j \in [n]$*, a transposition* $\tau(i,j)$ *leads to a new permutation obtained by swapping* π_i *and* π_j *in* $\boldsymbol{\pi}$*, i.e,*

$$\boldsymbol{\pi}\tau(i,j) = (\pi_1, \ldots, \pi_{i-1}, \pi_j, \pi_{i+1}, \ldots, \pi_{j-1}, \pi_i, \pi_{j+1}, \ldots, \pi_n).$$

If $|i-j| = 1$*,* $\tau(i,j)$ *is called the* adjacent transposition.

Definition 2. *For distinct* $i, j \in [n]$*, a translocation* $\phi(i,j)$ *leads to a new permutation obtained by moving* π_i *to the position of* j *and shifting symbols* $\pi_{[i+1,j]}$ *by one in* $\boldsymbol{\pi}$*. If* $i < j$*, there is*

$$\boldsymbol{\pi}\phi(i,j) = (\pi_1, \ldots, \pi_{i-1}, \pi_{i+1}, \ldots, \pi_j, \pi_i, \pi_{j+1}, \ldots, \pi_n).$$

In [15], Chee and Vu introduced the *generalized transposition* as follows. Denote $[a,b] \prec [c,d]$ as the interval $[a,b]$ precedes the interval $[c,d]$.

Definition 3. *For distinct two intervals* $A = [i,j], B = [k,\ell]$*, a generalized transposition* $\tau_g(A,B)$ *leads to a new permutation obtained by swapping* $\boldsymbol{\pi}_{[i,j]}$ *and* $\boldsymbol{\pi}_{[k,\ell]}$ *in* $\boldsymbol{\pi}$*. If* $A \prec B$*, there is*

$$\boldsymbol{\pi}\tau_g(A,B) = (\pi_1, \ldots, \pi_{i-1}, \pi_k, \pi_{k+1}, \ldots, \pi_\ell, \pi_{j+1}, \ldots, \pi_{k-1}, \pi_i, \ldots, \pi_j, \pi_{\ell+1}, \ldots, \pi_n).$$

If $k - j = 1$*,* $\tau_g(A,B)$ *is called the generalized adjacent transposition.*

We have $\phi(i,\ell) = \tau_g(A,B)$ with $A = [i,j], B = [k,\ell]$ if $i = j = k-1$. Hence, we say that a *translocation* can be considered as a special case of a *generalized adjacent transposition*.

Example 2. *Let* $\boldsymbol{\pi} = (1,6,4,3,2,5)$*. We have the following:*

$$\boldsymbol{\pi}\tau(2,5) = (1,2,4,3,6,5),$$
$$\boldsymbol{\pi}\phi(2,5) = (1,4,3,2,6,5),$$
$$\boldsymbol{\pi}\tau_g([2,3],[4,6]) = (1,3,2,5,6,4).$$

Definition 4. *The Hamming distance between two permutations* $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{S}_n$*, denoted by* $d_H(\boldsymbol{\pi}, \boldsymbol{\sigma})$*, is defined as the number of positions for which* $\boldsymbol{\pi}$ *and* $\boldsymbol{\sigma}$ *differ, i.e,*

$$d_H(\boldsymbol{\pi}, \boldsymbol{\sigma}) = |\{i \in [n] : \pi_i \neq \sigma_i\}|.$$

Definition 5. *The Levenshtein distance between two permutations* $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{S}_n$*, denoted by* $d_V(\boldsymbol{\pi}, \boldsymbol{\sigma})$*, is defined as the minimum number of insertions or deletions which are needed to change* $\boldsymbol{\pi}$ *to* $\boldsymbol{\sigma}$*.*

Definition 6. *The Kendall-tau distance between two permutations* $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{S}_n$*, denoted by* $d_K(\boldsymbol{\pi}, \boldsymbol{\sigma})$*, is defined as the minimum number of adjacent transpositions which are needed to change* $\boldsymbol{\pi}$ *to* $\boldsymbol{\sigma}$*, i.e,*

$$d_K(\boldsymbol{\pi}, \boldsymbol{\sigma}) = \min\{m : \boldsymbol{\pi}\tau_1\tau_2\cdots\tau_m = \boldsymbol{\sigma}\},$$

where $\tau_1, \ldots, \tau_m$ *are adjacent transpositions.*

Definition 7. *The generalized Kendall-tau distance between two permutations* $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{S}_n$*, denoted by* $d_{\bar{K}}(\boldsymbol{\pi}, \boldsymbol{\sigma})$*, is defined as the minimum number of generalized adjacent transpositions which are needed to change* $\boldsymbol{\pi}$ *to* $\boldsymbol{\sigma}$*, i.e,*

$$d_{\bar{K}}(\boldsymbol{\pi}, \boldsymbol{\sigma}) = \min\{m : \boldsymbol{\pi}\tau_{g1}\tau_{g2}\cdots\tau_{gm} = \boldsymbol{\sigma}\},$$

where $\tau_{g1}, \ldots, \tau_{gm}$ *are generalized adjacent transpositions.*

Definition 8. *The Ulam distance between two permutations* $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{S}_n$*, denoted by* $d_U(\boldsymbol{\pi}, \boldsymbol{\sigma})$*, is defined as the minimum number of translocations which are needed to change* $\boldsymbol{\pi}$ *to* $\boldsymbol{\sigma}$*, i.e,*

$$d_U(\boldsymbol{\pi}, \boldsymbol{\sigma}) = \min\{m : \boldsymbol{\pi}\phi_1\phi_2\cdots\phi_m = \boldsymbol{\sigma}\},$$

where $\phi_1, \ldots, \phi_m$ *are translocations.*

Proposition 1. *For two permutations* $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{S}_n$*, let* $\ell(\boldsymbol{\pi}, \boldsymbol{\sigma})$ *be the length of a longest common subsequence of* $\boldsymbol{\pi}$ *and* $\boldsymbol{\sigma}$*. The Ulam distance* $d_U(\boldsymbol{\pi}, \boldsymbol{\sigma})$ *between* $\boldsymbol{\pi}$ *and* $\boldsymbol{\sigma}$ *equals* $n - \ell(\boldsymbol{\pi}, \boldsymbol{\sigma})$*.*

Example 3. *Let* $\boldsymbol{\pi} = (4,3,1,2,5)$ *and* $\boldsymbol{\sigma} = (4,3,5,1,2)$*. Then* $d_H(\boldsymbol{\pi}, \boldsymbol{\sigma}) = 3$ *and* $d_U(\boldsymbol{\pi}, \boldsymbol{\sigma}) = 1$*.*

B. Permutation Code

At most t *deletions* deletes at most t symbols from the permutation $\boldsymbol{\pi}$, leading to $\boldsymbol{\pi}'$. $\mathcal{B}_t(\boldsymbol{\pi})$ is the set of all vectors of length at least $n-t$ received as a result of at most t deletions in $\boldsymbol{\pi}$. Our goal in this paper will be to design codes $\mathcal{P}_t(n)$ over $\mathcal{S}_n$ where for any $\boldsymbol{\pi} \in \mathcal{P}_t(n)$, we can recover $\boldsymbol{\pi}$ from $\boldsymbol{\pi}'$, provided that $\boldsymbol{\pi}'$ is the result of at most t deletions occurring in $\boldsymbol{\pi}$. The code $\mathcal{P}_t(n)$ is a t-deletion permutation code if and only if for distinct $\boldsymbol{\pi}_1, \boldsymbol{\pi}_2 \in \mathcal{P}_t(n)$ such that $\mathcal{B}_t(\boldsymbol{\pi}_1) \cap \mathcal{B}_t(\boldsymbol{\pi}_2) = \emptyset$.

The size of a permutation code $\mathcal{C} \subseteq \mathcal{S}_n$ is denoted $|\mathcal{C}|$ and its redundancy is defined as $\log(n!/|\mathcal{C}|)$. All logarithms in this paper are to the base 2.

Lemma 1. *Let* $n > t$ *be positive integers, the maximum size of a permutation code for correcting* t *deletions over* $\mathcal{S}_n$ *is at most* $n!/n^t$*.*

Proof: Let $\mathcal{B}_t(\mathcal{S}_n) = \cup_{\boldsymbol{\pi} \in \mathcal{S}_n} \mathcal{B}_t(\boldsymbol{\pi})$. It can be seen that $\mathcal{B}_t(\mathcal{S}_n)$ is the set of all sequences consisting of $n-t$ distinct symbols from an alphabet size n. Applying the sphere-packing method and the fact that $\binom{n}{t} \sim \frac{n^t}{t!}$ as $n \to \infty$, for fixed t, we have

$$\frac{n!}{t!} \geq |\mathcal{P}_t(n)| \cdot \binom{n}{t},$$
$$\Rightarrow \frac{n!}{t!} \gtrsim |\mathcal{P}_t(n)| \cdot \frac{n^t}{t!},$$
$$\Rightarrow |\mathcal{P}_t(n)| \lesssim \frac{n!}{n^t}.$$

This completes the proof. ■

Corollary 1. *The lower bound of the minimal redundancy of the permutation codes for correcting* t *deletions is* $t \log n$*.*

Lemma 2 (Section IV-C, [8]). *Let $\mathcal{C}_n$ be a permutation code for correcting t deletions with length n. The size of the code $\mathcal{C}_n$ is*

$$|\mathcal{C}_n| \geq \frac{(\frac{2n}{t}!)^{\frac{t}{2}}}{(n+1)^{3t/2-4}(\frac{2n}{t})^{\frac{t}{2}}}.$$

The redundancy of the code $\mathcal{C}_n$ proposed in [8] is far from the lower bound of the minimal redundancy of the permutation code for correcting t deletions $t \log n$.

III. Relationship of distances between two permutations

In this section, we first build a tight inequality relationship between the Ulam distance and Hamming distance via a novel mapping function. Furthermore, due to the close relationship between Ulam distance and Levenshtein/generalized Kendall-tau distance, we also provide the connection between Levenshtein/generalized Kendall-tau distance and Hamming distance in this section. Hence, we depict a big picture to connect different kinds of distances such that we can construct permutation codes for deletions/transpositions/translocations by the well-studied permutation codes for substitutions in the Hamming metric.

In [16], Levenshtein demonstrated that for sequences $\boldsymbol{x}$ and $\boldsymbol{y}$ with a length of n, $d_V(\boldsymbol{x}, \boldsymbol{y}) = 2(n - \ell(\boldsymbol{x}, \boldsymbol{y}))$. This formula is applicable to permutations as well. Combining with the Definition 1, we have:

$$d_V(\boldsymbol{\pi}, \boldsymbol{\sigma}) = 2 d_U(\boldsymbol{\pi}, \boldsymbol{\sigma}). \tag{1}$$

In [6], Farnoud et al. derived inequalities to present the relationship between the Ulam distance and Hamming distance over two permutations $\boldsymbol{\pi}$ and $\boldsymbol{\sigma}$ as follows:

$$\frac{1}{n} d_H(\boldsymbol{\pi}, \boldsymbol{\sigma}) \leq d_U(\boldsymbol{\pi}, \boldsymbol{\sigma}) \leq d_H(\boldsymbol{\pi}, \boldsymbol{\sigma}). \tag{2}$$

In this paper, we propose a novel bijective mapping function such that we can obtain a tighter inequality relationship between the Ulam distance and Hamming distance over two permutations.

For convenience, we first append $n+1$ at the end of $\boldsymbol{\pi}$ and denote it as $\bar{\boldsymbol{\pi}} = (\boldsymbol{\pi}, n+1)$. We define the subset $\mathcal{S}'_{n+1}$ of $\mathcal{S}_{n+1}$ as follows:

$$\mathcal{S}'_{n+1} = \{\bar{\boldsymbol{\pi}} \in \mathcal{S}_{n+1} \mid \bar{\boldsymbol{\pi}}_{n+1} = n+1\}$$

This subset consists of all permutations in $\mathcal{S}_{n+1}$ where the last element is fixed as $n+1$.

Definition 9. *Given a permutation $\boldsymbol{\pi} \in \mathcal{S}_n$, we define the function $\boldsymbol{f} : \mathcal{S}'_{n+1} \to f[\mathcal{S}'_{n+1}]$, where $f[\mathcal{S}'_{n+1}] = \{f(\bar{\boldsymbol{\pi}}) : \bar{\boldsymbol{\pi}} \in \mathcal{S}'_{n+1}\}$, such that:*

$$f(\bar{\boldsymbol{\pi}})_i = \begin{cases} \bar{\pi}_{\bar{\pi}_i^{-1}+1}, & i = 1, 2, \ldots, n \\ \bar{\pi}_1, & i = n+1 \end{cases} \tag{3}$$

Lemma 3. *The function $\boldsymbol{f} : \mathcal{S}'_{n+1} \to f[\mathcal{S}'_{n+1}]$ is bijective.*

Proof: To prove that $\boldsymbol{f}$ is bijective, we must demonstrate its injectivity and subjectivity. Assume we have two permutations $\bar{\boldsymbol{\pi}}$ and $\bar{\boldsymbol{\sigma}}$ in $\mathcal{S}_{n+1}$ such that $\boldsymbol{f}(\bar{\boldsymbol{\pi}}) = \boldsymbol{f}(\bar{\boldsymbol{\sigma}})$. This implies: $f(\bar{\boldsymbol{\pi}})_i = f(\bar{\boldsymbol{\sigma}})_i$, for all $i = 1, 2, \ldots, n+1$. From the definition of f, we have two cases:

- For $i = 1, 2, \ldots, n$, $f(\bar{\boldsymbol{\pi}})_i = \bar{\pi}_{\bar{\pi}_i^{-1}+1} = f(\bar{\boldsymbol{\sigma}})_i = \bar{\sigma}_{\bar{\sigma}_i^{-1}+1}$.
- For $i = n+1$, $f(\bar{\boldsymbol{\pi}})_{n+1} = \bar{\pi}_1 = f(\bar{\boldsymbol{\sigma}})_{n+1} = \bar{\sigma}_1$.

Since $\boldsymbol{f}(\bar{\boldsymbol{\pi}}) = \boldsymbol{f}(\bar{\boldsymbol{\sigma}})$ maps each element i in the permutation $\bar{\boldsymbol{\pi}}$ to the element that follows it, and the last element to the first, it implies both $\bar{\boldsymbol{\pi}}$ and $\bar{\boldsymbol{\sigma}}$ are identical sequences. Hence, $\bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\sigma}}$, proving infectivity.

By the definition of the function $\boldsymbol{f}$, the function $\boldsymbol{f}$ is also subjective. Hence, having shown both injectivity and surjectivity, we conclude that the mapping function $\boldsymbol{f}$ is bijective. ∎

Given the definitions of $\mathcal{S}'_{n+1}$ as the subset of permutations in $\mathcal{S}_{n+1}$ where each permutation specifically ends with the element $n+1$, and $f[\mathcal{S}'_{n+1}]$ as the image of $\mathcal{S}'_{n+1}$ under the mapping f, an important observation regarding their sizes can be made. The subset $\mathcal{S}'_{n+1}$ consists of all possible arrangements of the first n elements since the last element is fixed. Therefore, the size of $\mathcal{S}'_{n+1}$ is determined by the number of permutations of these n elements, which is exactly $n!$.

Furthermore, the function f is defined as a one-to-one mapping from $\mathcal{S}'_{n+1}$ onto its image within $\mathcal{S}_{n+1}$. This injective property ensures that for each permutation in $\mathcal{S}'_{n+1}$, there is a unique permutation in the image $f[\mathcal{S}'_{n+1}]$, thereby preserving the size of the image as $n!$.

Then, for any given $\boldsymbol{\sigma} \in f[\mathcal{S}'_{n+1}]$, we denote the inverse function $\boldsymbol{f}^{-1}(\cdot) : f[\mathcal{S}'_{n+1}] \to \mathcal{S}'_{n+1}$ as follows:

$$f^{-1}(\boldsymbol{\sigma})_i = \begin{cases} \sigma_{n+1}, & i = 1 \\ \sigma_{f^{-1}(\boldsymbol{\sigma})_{i-1}}, & i = 2, 3, \ldots, n+1 \end{cases} \tag{4}$$

Example 4. *Suppose $\boldsymbol{\pi} = (1, 3, 4, 2, 5) \in \mathcal{S}_5$, we have $\bar{\boldsymbol{\pi}} = (1, 3, 4, 2, 5, 6) \in \mathcal{S}_5$ and $\bar{\boldsymbol{\pi}}^{-1} = (1, 4, 2, 3, 5, 6)$. Hence, $f(\bar{\boldsymbol{\pi}}) = (3, 5, 4, 2, 6, 1)$.*

After introducing the mapping function $f(\cdot)$, we can build a relationship between Ulam distance over $\boldsymbol{\pi}$ and $\boldsymbol{\sigma}$ and Hamming distance over two permutations $f(\boldsymbol{\pi})$ and $f(\boldsymbol{\sigma})$ given $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{S}_n$.

Theorem 1. *Given two permutations $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{S}_n$, we have $d_U(\boldsymbol{\pi}, \boldsymbol{\sigma}) \geq \frac{1}{3} d_H(f(\bar{\boldsymbol{\pi}}), f(\bar{\boldsymbol{\sigma}}))$.*

Proof: Suppose there is a translocation between $\boldsymbol{\pi}$ and $\boldsymbol{\sigma}$, we denote as $\phi(i, j)$ and $i < j < n$, i.e., $\boldsymbol{\pi}\phi(i, j) = \boldsymbol{\sigma}$. We have subsequences $\boldsymbol{\pi}_{[i,j]}$ and $\boldsymbol{\sigma}_{[i,j]}$ in $\boldsymbol{\pi}$ and $\boldsymbol{\sigma}$, respectively as follows:

$$\begin{aligned} \boldsymbol{\pi}_{[i,j]} &= (\pi_i, \pi_{i+1}, \ldots, \pi_j) \\ \boldsymbol{\sigma}_{[i,j]} &= (\sigma_i, \sigma_{i+1}, \ldots, \sigma_j) \end{aligned} \tag{5}$$

From the definition of translocation $\phi(i, j)$, (5) can be rewritten as:

$$\begin{aligned} \boldsymbol{\pi}_{[i-1,j+1]} &= (\pi_{i-1}, \pi_i, \pi_{i+1}, \ldots, \pi_{j-1}, \pi_j, \pi_{j+1}) \\ \boldsymbol{\sigma}_{[i-1,j+1]} &= (\pi_{i-1}, \pi_{i+1}, \pi_{i+2}, \ldots, \pi_j, \pi_i, \pi_{j+1}) \end{aligned}$$

Since $i < j < n$, we have $\boldsymbol{\pi}_{[i-1,j+1]} = \bar{\boldsymbol{\pi}}_{[i-1,j+1]}$ and $\boldsymbol{\sigma}_{[i-1,j+1]} = \bar{\boldsymbol{\sigma}}_{[i-1,j+1]}$. Let us revisit the definition of the mapping function $f(\cdot)$, we can find the following changes:

$$f(\bar{\boldsymbol{\pi}})_{\bar{\pi}_{i-1}} = \pi_i \to f(\bar{\boldsymbol{\sigma}})_{\bar{\pi}_{i-1}} = \pi_{i+1}$$
$$f(\bar{\boldsymbol{\pi}})_{\bar{\pi}_i} = \pi_{i+1} \to f(\bar{\boldsymbol{\sigma}})_{\bar{\pi}_i} = \pi_{j+1}$$
$$f(\bar{\boldsymbol{\pi}})_{\bar{\pi}_j} = \pi_{j+1} \to f(\bar{\boldsymbol{\sigma}})_{\bar{\pi}_j} = \pi_i$$

Besides, since the order of the elements in the interval $[i+1, j-1]$ does not change, it means

$$f(\bar{\boldsymbol{\pi}})_{[\bar{\pi}_{i+1},\bar{\pi}_{j-1}]} = f(\bar{\boldsymbol{\sigma}})_{[\bar{\pi}_{i+1},\bar{\pi}_{j-1}]}.$$

Therefore, each translocation between $\boldsymbol{\pi}$ and $\boldsymbol{\sigma}$ can be viewed as 3 substitutions between $f(\boldsymbol{\pi})$ and $f(\boldsymbol{\sigma})$, implying that

$$d_H(f(\bar{\boldsymbol{\pi}}), f(\bar{\boldsymbol{\sigma}})) \leq 3d_U(\boldsymbol{\pi}, \boldsymbol{\sigma}). \blacksquare$$

Example 5. *Suppose* $\boldsymbol{\pi} = (1,3,4,2,5)$, $\boldsymbol{\sigma} = (4,2,3,5,1)$ *and* $d_U(\boldsymbol{\pi},\boldsymbol{\sigma}) = 2$, *we have* $\bar{\boldsymbol{\pi}} = (1,3,4,2,5,6)$ *and* $\bar{\boldsymbol{\sigma}} = (4,2,3,5,1,6)$. *Then, we have* $f(\bar{\boldsymbol{\pi}}) = (3,5,4,2,6,1)$ *and* $f(\bar{\boldsymbol{\sigma}}) = (6,3,5,2,1,4)$. *Hence* $d_H(f(\bar{\boldsymbol{\pi}}), f(\bar{\boldsymbol{\sigma}})) = 5 \leq 3d_U(\boldsymbol{\pi},\boldsymbol{\sigma})$.

Then, we will show the permutation codes for correcting t deletions with the help of Theorem 1 and some known permutation codes in the Hamming metric.

Lemma 4 (Theorem 7, [8])**.** *The permutation code* $\mathcal{C} \in \mathcal{S}_n$ *is capable of correcting* t *deletions if and only if* $d_U(\boldsymbol{\pi},\boldsymbol{\sigma}) \geq t+1$, *for all* $\boldsymbol{\pi},\boldsymbol{\sigma} \in \mathcal{C}$, $\boldsymbol{\pi} \neq \boldsymbol{\sigma}$.

Theorem 2. *Let* $\mathcal{C} \in \mathcal{S}_n$ *be a permutation code where* $d_H(\boldsymbol{f}(\bar{\boldsymbol{\pi}}), \boldsymbol{f}(\bar{\boldsymbol{\sigma}})) \geq 3(t+1)$, *for all* $\boldsymbol{\pi},\boldsymbol{\sigma} \in \mathcal{C}$, $\boldsymbol{\pi} \neq \boldsymbol{\sigma}$. *Then,* $\mathcal{C}$ *is capable of correcting* t *deletions.*

Proof: From Theorem 1, $d_H(\boldsymbol{f}(\bar{\boldsymbol{\pi}}), \boldsymbol{f}(\bar{\boldsymbol{\sigma}})) \geq 3(t+1)$ implies $d_U(\boldsymbol{\pi},\boldsymbol{\sigma}) \geq t+1$. Hence, this statement holds as Lemma 4. ■

The construction of permutation codes in the Hamming metric is well-studied in [17]–[19]. Here, we apply the permutation code proposed in the Hamming metric in [18] as the base code.

Lemma 5. *(Theorem 2, [18]) Let* $\mathcal{P}_H(n,d)$ *denote the permutation codes of length* n *with minimum Hamming distance* d *and* $M(n,d)$ *be the minimum size of the permutation code* $\mathcal{P}_H(n,d)$. *For two integers* d, n *with* $4 \leq d \leq n$, *one has*

$$M(n,d) \geq \frac{n!}{p^{d-2}},$$

where p *is the smallest prime larger than or equal to* n.

From Theorem 2 and Lemma 5, we immediately have the following corollary.

Corollary 2. *There exists a permutation code with length* n *that is capable of correcting* t *deletions with the code size at least* $\frac{n!}{p^{3t+1}}$, *where* p *is the smallest prime bigger than or equal to* n. *Hence, it follows that the redundancy of this permutation code is at most* $(3t+1)\log n + o(\log n)$ *bits.*

Furthermore, we can build the following relationship between Levenshtein distance and Hamming distance over two permutations.

Corollary 3. *Given two permutations* $\boldsymbol{\pi},\boldsymbol{\sigma} \in \mathcal{S}_n$, *we have* $d_H(f(\bar{\boldsymbol{\pi}}), f(\bar{\boldsymbol{\sigma}})) \leq \frac{3}{2}d_V(\boldsymbol{\pi},\boldsymbol{\sigma})$.

Based on the definition, it apparently follows that:

$$d_{\bar{K}}(\boldsymbol{\pi},\boldsymbol{\sigma}) \leq d_U(\boldsymbol{\pi},\boldsymbol{\sigma}) \leq d_K(\boldsymbol{\pi},\boldsymbol{\sigma}).$$

Therefore, a natural consequence from Theorem 1 is to build the relationship between generalized Kendall-tau distance and Hamming distance.

Corollary 4. *Given two permutations* $\boldsymbol{\pi},\boldsymbol{\sigma} \in \mathcal{S}_n$, *we have*

$$d_H(f(\bar{\boldsymbol{\pi}}), f(\bar{\boldsymbol{\sigma}})) \leq 3d_{\bar{K}}(\boldsymbol{\pi},\boldsymbol{\sigma})$$

Proof: The idea of this proof is almost identical to the proof of Theorem 1. ■

Consequently, based on Theorem 1 and its corollaries mentioned above, we simplify the complex task of developing permutation codes in the Levenshtein, Ulam, and generalized Kendall-tau metrics by transforming it into the construction of permutation codes in the Hamming metric, an area that has been extensively studied.

IV. Codes Construction

In this section, we present our construction of permutation codes for correcting t deletions with a specific decoding process. Different from the permutation codes for correcting t deletions proposed in [8] relying on the auxiliary codes in the Ulam metric, our construction is achieved by incorporating the base code in the Hamming metric.

Suppose t deletions occur in $\boldsymbol{\pi} \in \mathcal{S}_n$, we denote the received sequence as $\boldsymbol{\pi}' = (\pi'_1, \pi'_2, \ldots, \pi'_{n-t})$ and the t deleted symbols as $\boldsymbol{\pi}'' = (\pi''_1, \pi''_2, \ldots, \pi''_t)$.

Then, we append $\boldsymbol{\pi}''$ at the end of $(\boldsymbol{\pi}', n+1)$ and let $\hat{\boldsymbol{\pi}} = (\boldsymbol{\pi}', n+1, \boldsymbol{\pi}'')$.

Lemma 6. *Suppose* t *deletions occur in* $\boldsymbol{\pi} \in \mathcal{S}_n$, *we denote the received sequence as* $\boldsymbol{\pi}' = (\pi'_1, \pi'_2, \ldots, \pi'_{n-t})$ *and the* t *deleted symbols as* $\boldsymbol{\pi}'' = (\pi''_1, \pi''_2, \ldots, \pi''_t)$. *Append* $\boldsymbol{\pi}''$ *at the end of* $(\boldsymbol{\pi}', n+1)$ *and let* $\hat{\boldsymbol{\pi}} = (\boldsymbol{\pi}', n+1, \boldsymbol{\pi}'')$. *We have* $d_H(f(\bar{\boldsymbol{\pi}}), f(\hat{\boldsymbol{\pi}})) \leq 2t+1$.

Proof: $\bar{\boldsymbol{\pi}}$ and $\hat{\boldsymbol{\pi}}$ can be written as follows, respectively:

$$\bar{\boldsymbol{\pi}} = (\pi_1, \cdots, \pi_{i-1}, \pi_i, \pi_{i+1}, \cdots, n+1)$$
$$\hat{\boldsymbol{\pi}} = (\pi'_1, \pi'_2, \cdots, \pi'_{n-t}, n+1, \pi''_1, \pi''_2, \cdots, \pi''_t)$$

We first give toy examples when a single deletion or two deletions occur. Suppose there is only a single deletion, i.e., π_i is deleted from $\boldsymbol{\pi}$. The corresponding $\bar{\boldsymbol{\pi}}$ and $\hat{\boldsymbol{\pi}}$ can be shown as follows:

$$\bar{\boldsymbol{\pi}} = (\pi_1, \cdots, \pi_{i-1}, \pi_i, \pi_{i+1}, \pi_{i+2}, \cdots, n+1)$$
$$\hat{\boldsymbol{\pi}} = (\pi_1, \cdots, \pi_{i-1}, \pi_{i+1}, \pi_{i+2} \cdots, n+1, \pi_i) \quad (6)$$

From (6), we have

$$f(\bar{\boldsymbol{\pi}})_{\pi_{i-1}} \neq f(\hat{\boldsymbol{\pi}})_{\pi_{i-1}},$$

$$f(\bar{\boldsymbol{\pi}})_{\pi_i} \neq f(\hat{\boldsymbol{\pi}})_{\pi_i},$$
$$f(\bar{\boldsymbol{\pi}})_{n+1} \neq f(\hat{\boldsymbol{\pi}})_{n+1},$$

and all other elements in $f(\bar{\boldsymbol{\pi}})$ and $f(\hat{\boldsymbol{\pi}})$ are identical based on the definition of the mapping function $f(\cdot)$.

Next, suppose there are two deletions, i.e., π_i and π_j are deleted from $\boldsymbol{\pi}$ with $i < j$. The corresponding $\bar{\boldsymbol{\pi}}$ and $\hat{\boldsymbol{\pi}}$ can be shown as follows:

$$\bar{\boldsymbol{\pi}} = (\pi_1, \cdots, \pi_{i-1}, \pi_i, \pi_{i+1}, \cdots, \pi_{j-1}, \pi_j, \pi_{j+1}, \cdots, n+1)$$
$$\hat{\boldsymbol{\pi}} = (\pi_1, \cdots, \pi_{i-1}, \pi_{i+1}, \cdots, \pi_{j-1}, \pi_{j+1}, \cdots, n+1, \pi_i, \pi_j).$$

Similarly, if $j \neq i+1$ (π_i and π_j is not adjacent in $\boldsymbol{\pi}$). We have

$$f(\bar{\boldsymbol{\pi}})_{\pi_{i-1}} \neq f(\hat{\boldsymbol{\pi}})_{\pi_{i-1}}, f(\bar{\boldsymbol{\pi}})_{\pi_i} \neq f(\hat{\boldsymbol{\pi}})_{\pi_i},$$
$$f(\bar{\boldsymbol{\pi}})_{\pi_{j-1}} \neq f(\hat{\boldsymbol{\pi}})_{\pi_{j-1}}, f(\bar{\boldsymbol{\pi}})_{\pi_j} \neq f(\hat{\boldsymbol{\pi}})_{\pi_j},$$
$$f(\bar{\boldsymbol{\pi}})_{n+1} \neq f(\hat{\boldsymbol{\pi}})_{n+1},$$

and all other elements in $f(\bar{\boldsymbol{\pi}})$ and $f(\hat{\boldsymbol{\pi}})$ are identical.

Let us go to the general t deletions. Suppose a deletion π_u in $\boldsymbol{\pi}$ and it corresponds π''_v in $\boldsymbol{\pi}''$. We can see that

$$f(\bar{\boldsymbol{\pi}})_{\pi_{u-1}} \neq f(\hat{\boldsymbol{\pi}})_{\pi_{u-1}},$$
$$f(\bar{\boldsymbol{\pi}})_{\pi_u} \neq f(\hat{\boldsymbol{\pi}})_{\pi_u},$$
$$f(\bar{\boldsymbol{\pi}})_{\pi''_{v-1}} \neq f(\hat{\boldsymbol{\pi}})_{\pi''_{v-1}}.$$

It is worth noting that $f(\bar{\boldsymbol{\pi}})_{\pi''_{v-1}} \neq f(\hat{\boldsymbol{\pi}})_{\pi''_{v-1}}$ also means $f(\bar{\boldsymbol{\pi}})_{\pi_r} \neq f(\hat{\boldsymbol{\pi}})_{\pi_r}$, where π_r is another deletion in $\boldsymbol{\pi}$ and corresponds π''_{v-1} in $\boldsymbol{\pi}''$. It also means this difference has been counted before and we only need to count $f(\bar{\boldsymbol{\pi}})_{n+1} \neq f(\hat{\boldsymbol{\pi}})_{n+1}$ here. Hence, each deletion in $\boldsymbol{\pi}$ refers to two substitutions between $f(\bar{\boldsymbol{\pi}})$ and $f(\hat{\boldsymbol{\pi}})$. Besides, there is another substitution between $f(\bar{\boldsymbol{\pi}})_{n+1}$ and $f(\hat{\boldsymbol{\pi}})_{n+1}$.

Overall, we can conclude that $d_H(f(\bar{\boldsymbol{\pi}}), f(\hat{\boldsymbol{\pi}})) \leq 2t+1$. ■

Example 6. *Let $\boldsymbol{\pi} = (1,3,4,2,5,6,9,8,7) \in \mathcal{S}_9$ and $t = 3$ with $(3,2,8)$ are deleted. Then, we have the following:*

$$\bar{\boldsymbol{\pi}} = (1,3,4,2,5,6,9,8,7,10),$$
$$f(\bar{\boldsymbol{\pi}}) = (3,5,4,2,6,9,10,7,8,1)$$

$$\hat{\boldsymbol{\pi}} = (1,4,5,6,9,7,10,2,3,8),$$
$$f(\hat{\boldsymbol{\pi}}) = (\underline{4},\underline{3},\underline{8},\underline{5},6,9,10,\underline{1},\underline{7},\underline{2}),$$

where the symbol with underline denotes the symbol is substituted. We can see that $d_H(f(\bar{\boldsymbol{\pi}}), f(\hat{\boldsymbol{\pi}})) = 7 \leq 2t+1$ when $t = 3$.

Therefore, we only need to correct substitutions in permutations here. Then, we are going to apply the code proposed in [18] to correct these $2t+1$ substitutions in $f(\bar{\boldsymbol{\pi}})$. Utilizing the permutation code $\mathcal{P}_H(n,d)$ in Lemma 5 as the base code, our proposed permutation codes for correcting t deletions are constructed as follows.

Theorem 3. *For two integers t, n, the permutation code*

$$\mathcal{P}_t(n) = \{\boldsymbol{\pi} \in \mathcal{S}_n : \boldsymbol{f}(\bar{\boldsymbol{\pi}}) \in \mathcal{P}_H(n+1, 4t+3)\}$$

is capable of correcting t deletions in $\boldsymbol{\pi}$ with $(4t+1)\log n + o(\log n)$ bits of redundancy.

Proof: At the receiver, we append the deleted symbols at the end of the received sequence and denote it as $\hat{\boldsymbol{\pi}}$. From Lemma 6, we have $d_H(\boldsymbol{f}(\bar{\boldsymbol{\pi}}), \boldsymbol{f}(\hat{\boldsymbol{\pi}})) \leq 2t+1$. Since $\boldsymbol{f}(\bar{\boldsymbol{\pi}}) \in \mathcal{P}_H(n+1, 4t+3)$, then $d_H(\boldsymbol{f}(\bar{\boldsymbol{\pi}}), \boldsymbol{f}(\bar{\boldsymbol{\sigma}})) \geq 4t+3$ for any two distinct codes $\boldsymbol{\pi}, \boldsymbol{\sigma} \in \mathcal{P}_t(n)$. Hence, it follows that $\boldsymbol{f}(\hat{\boldsymbol{\pi}}) \cap \boldsymbol{f}(\bar{\boldsymbol{\pi}}) = \emptyset$. Then, we can recover $\boldsymbol{f}(\bar{\boldsymbol{\pi}})$ via $\boldsymbol{f}(\hat{\boldsymbol{\pi}})$ and recover $\boldsymbol{\pi}$ via (4).

As for the redundancy, $\mathcal{P}_H(n+1, 4t+3) \cap f[\mathcal{S}'_{n+1}]$ can be considered as one of the sets among p^{4t+1} sets partitioned by the $f[\mathcal{S}'_{n+1}]$. Hence, we have

$$|\mathcal{P}_t(n)| \geq \frac{n!}{p^{4t+1}},$$

where p is the smallest prime larger than n. Therefore, we show that the redundancy is at most $(4t+1)\log n + o(\log n)$ bits. ■

It is worth noting that the code size of our proposed permutation codes for correcting t deletions greatly improves the results in [8] as shown in Lemma 2. Besides, the redundancy of our proposed code $\mathcal{P}_t(n)$ achieves the almost order optimal redundancy as $O(t \log n)$.

We denote the decoder of the code $\mathcal{P}_H(n,d)$ as $\mathcal{D}_H(n,d)$. The decoding process of $\mathcal{P}_t(n)$ can be summarized as Algorithm 1.

Algorithm 1: Decoding Algorithm of $\mathcal{P}_t(n)$

Input: Received Permutation $\boldsymbol{\pi}'$ after t deletions
Output: Originial Permuation $\boldsymbol{\pi} \in \mathcal{P}_t(n)$

Comparing $\boldsymbol{\pi}'$ and $[n]$, find the t deleted symbols and denote as $\boldsymbol{\pi}'' = (\pi''_1, \pi''_2, \ldots, \pi''_t)$.
$\hat{\boldsymbol{\pi}} \leftarrow (\boldsymbol{\pi}', n+1, \boldsymbol{\pi}'')$.
Obtain $f(\hat{\boldsymbol{\pi}})$ and recover $f(\bar{\boldsymbol{\pi}})$ via $\mathcal{D}_H(n, 4t+3)$.
Recover $\boldsymbol{\pi}$ from $f(\bar{\boldsymbol{\pi}})$ via the inverse mapping function $f^{-1}(\cdot)$.

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